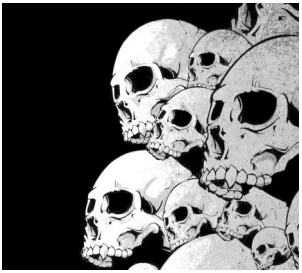


Y. Collette (ycollette.nospam@free.fr)
<https://audinux.github.io>

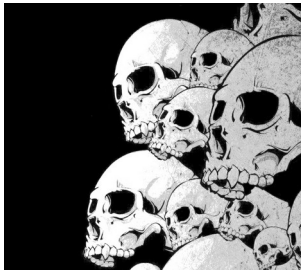




Guitarix

<https://guitarix.org/>





Some old promises

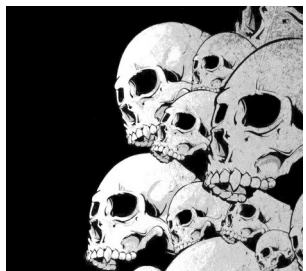
In the 1980s, the "death of tubes" was heralded with transistor distortions, the Rockman, and later the SansAmp—a promised revolution, but tube amps never disappeared.

In the 2000s and 2010s, Line 6 PODs, then Kemper/Axe-FX, were heralded as "the death of the amp and pedalboard"—again, they mostly added a new palette of tools.

Today, with NAM, AIDA-X, Neural DSP, etc., the narrative is back: "plugins/AI replace everything"... but amps, pedals, and open-source solutions like Guitarix + IR + LADSPA/LV2 under Linux continue to offer excellent sounds.

“For 40 years, every new technology has promised to kill tubes and pedals... Yet, the good sounds were already there. What’s really changing is the accessibility and diversity of tools.”

“No need to wait for the next revolution: with Guitarix, IR and free plugins under Linux, we already have the tools to sculpt a professional sound.”



Guitarix

General appearance

Activation of
different
windows

Effects
available

General
parameters

Tuner

Effect rack

Presets

23/05/26





Guitarix

The buttons bar

Pool: activates / deactivate the display of the effect list bar.

Order: Reduces the height of all effect modules and allows you to easily click and reorganize these effects. This view is practical to reorganize large amounts of effects.

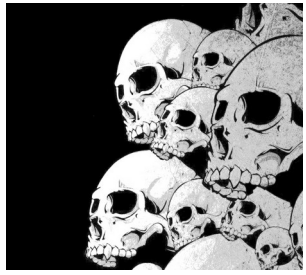
Tuner: Activates / deactivate the tacking of the tuning rack.

Rack: Activates / deactivate the Display of Effect Modules Rack.

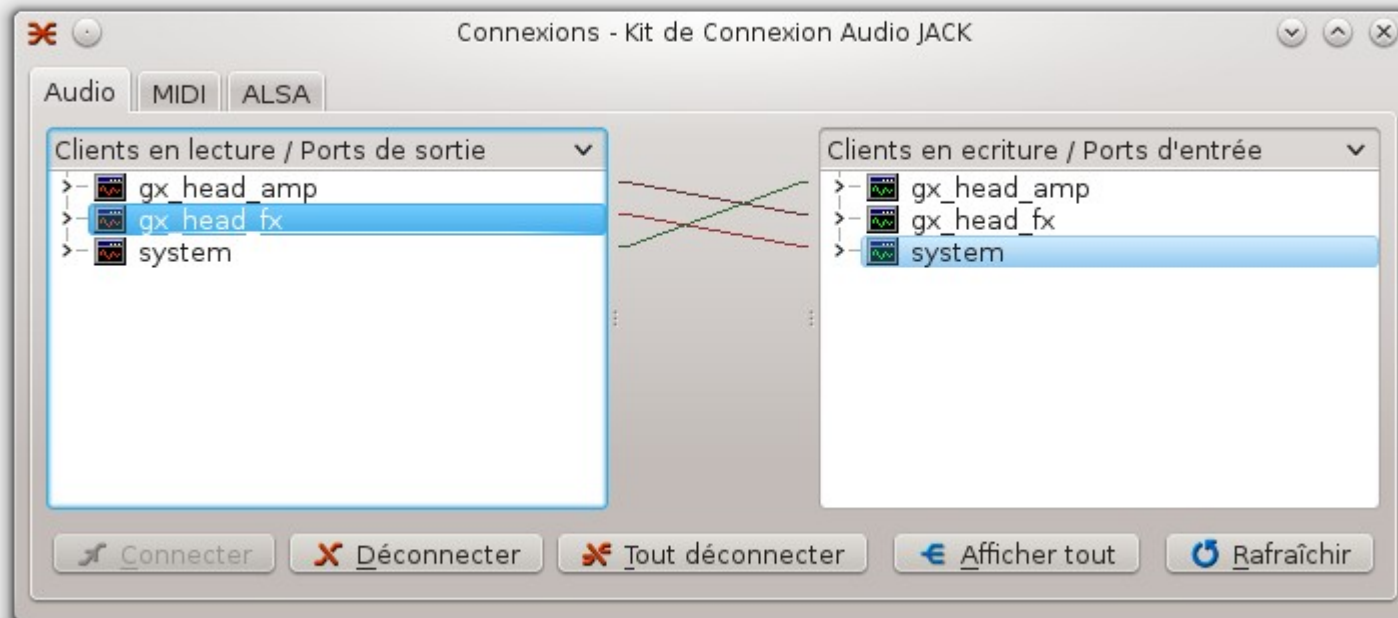
Split: displays the modules rack in two columns, mono grafts on the left column and stereo grafts on the right column.

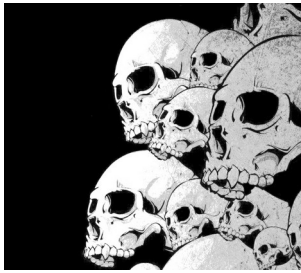
Fold / Show: the button less reduces the height of all effect modules. The button plus dilates.

Live: Displays a minimum full screen poster with preset information. Practical for live performance.



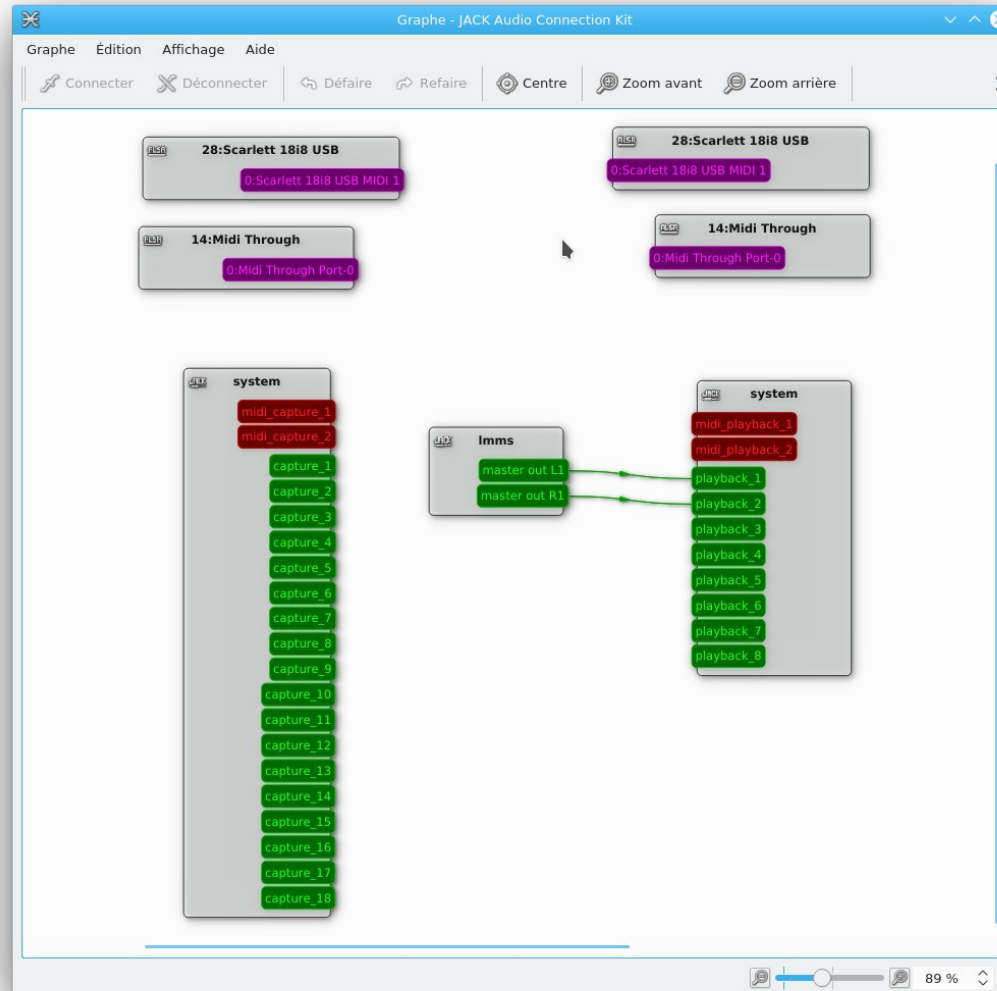
Guitarix Jack side





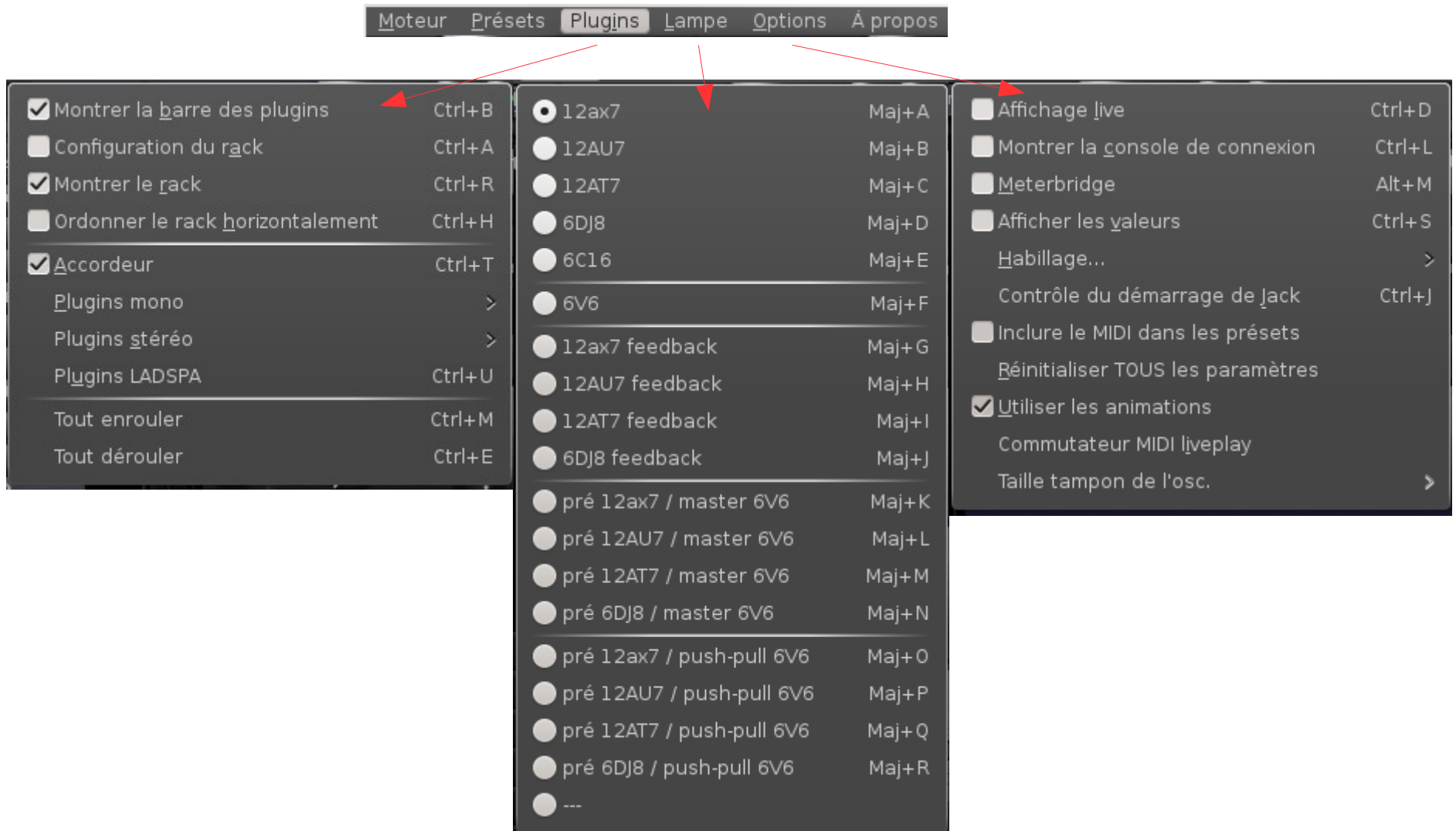
Guitarix

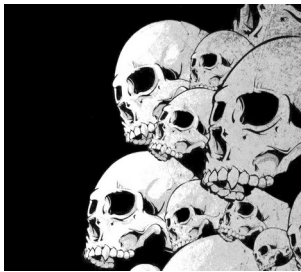
Jack side - New connection window



Guitarix

The Menus





Guitarix

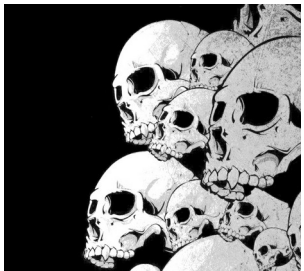
The structure of a rack

Pre-amplifier effects

Amplifier

Post-amplifier effects





The tube amp emulator



Mixture between signal with and without distortion

Signal gain

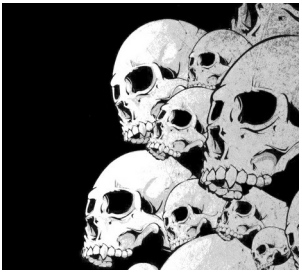
Sound characteristics settings

General volume

Setting with lamp choice just for the amplifier

12ax7
12AU7
12AT7
6DJ8
6C16
6V6
12ax7 feedback
12AU7 feedback
12AT7 feedback
6DJ8 feedback
pré 12ax7 / master 6V6
pré 12AU7 / master 6V6
pré 12AT7 / master 6V6
pré 6DJ8 / master 6V6
pré 12ax7 / push-pull 6V6
pré 12AU7 / push-pull 6V6
pré 12AT7 / push-pull 6V6
pré 6DJ8 / push-pull 6V6

Setting with choice of lamp for pre-amplifier / amplifier



Cabinet



A cabinet is a baffle simulator.
It allows you to simulate different types of
amplifier or a speaker combination.

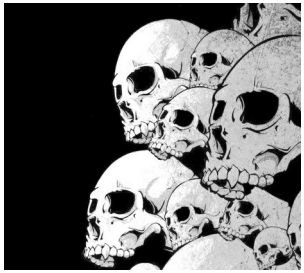
- 4X12
- 2X12
- 1X12
- 4X10
- 2X10
- HighGain Style
- Twin Style
- Bassman Style
- Marshall Style
- AC-30 Style
- Princeton Style
- A2 Style
- 1X15
- Mesa Boogie Style
- Brilliant
- Vitalize
- Charisma



Cabinet 4x12



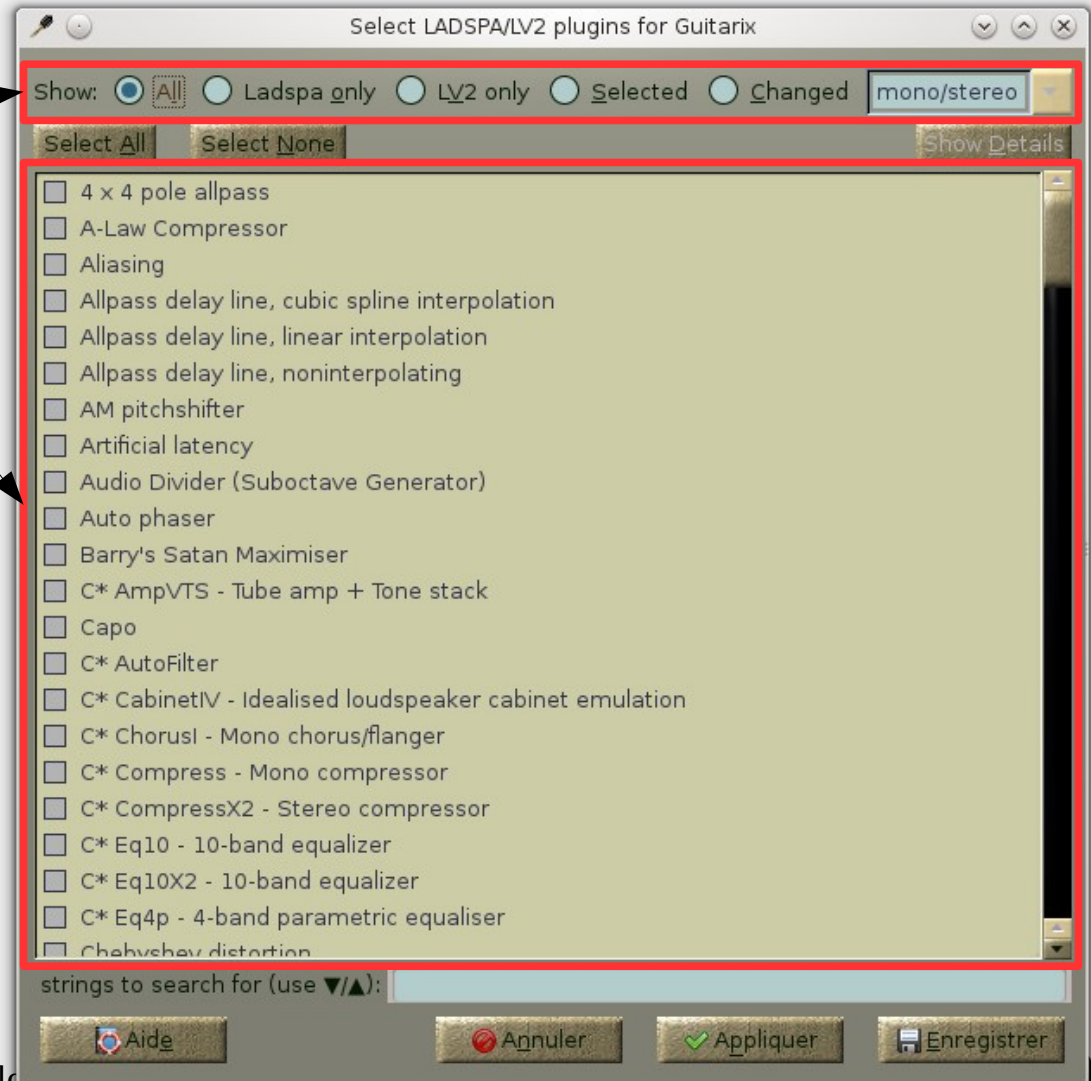
Cabinet Princeton style

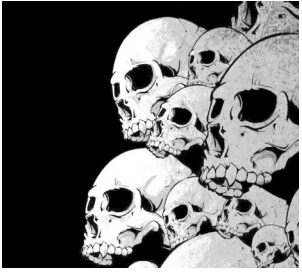


Plugin management LADSPA / LV2

Plugin type selection area
(LADSPA / LV2)

List and selection of plugins
to add to rack





Plugins for guitar

Interesting plugins for the guitar:

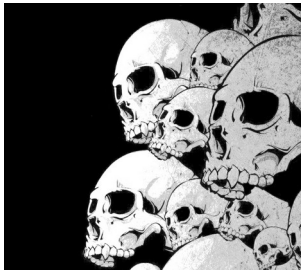
<https://portalmod.com>

A distortion in LV2 format from Portalmod:

```
$ git clone https://github.com/portalmod/mod-distortion.git
$ cd mod-distortion
$ make # compilation
$ make install # Installation in the user directory
$ sudo dnf install mod-distortion
```

Tonality pedals in LV2 format from Portalmode:

```
$ git clone https://github.com/portalmod/mod-pitchshifter.git
$ cd mod-pitchshifter
$ make # compilation
$ make install # Installation in the user directory
$ sudo dnf install mod-gxpitchshifter
```

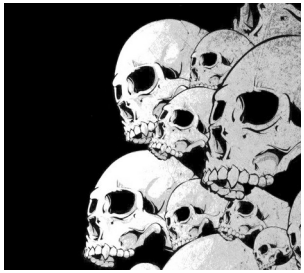


Check via MIDI

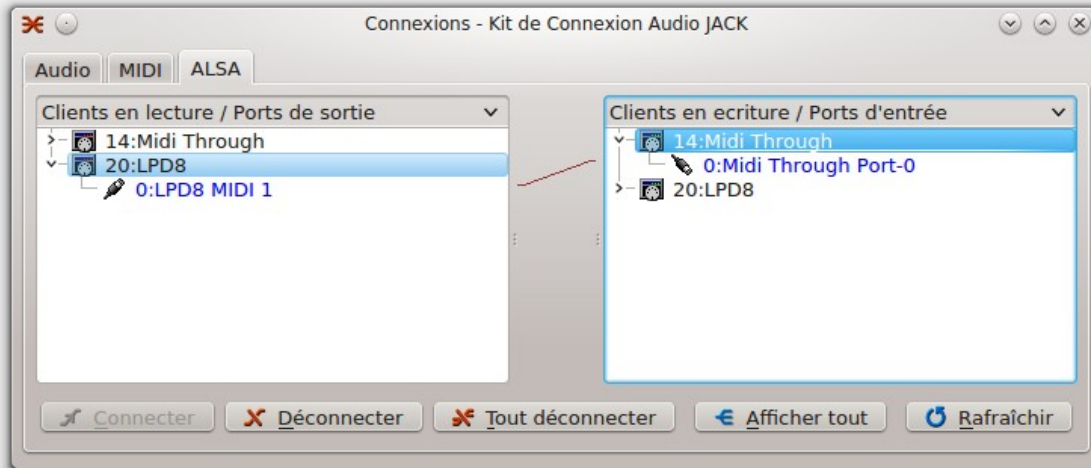


Objective: control an effect button via a MIDI instrument



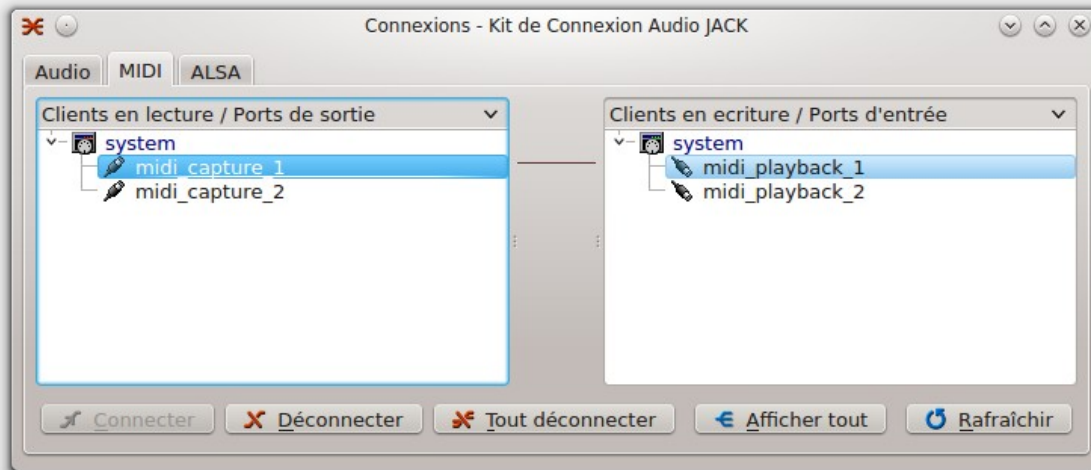


Control via Midi Jack connections



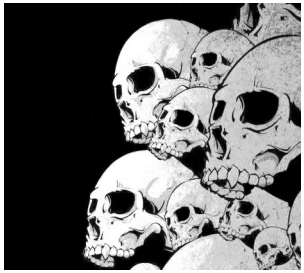
First step:
Alsa tab:

We connect the MIDI output of
the LDP8 to the System system
input: MIDI_PLAYBACK_1



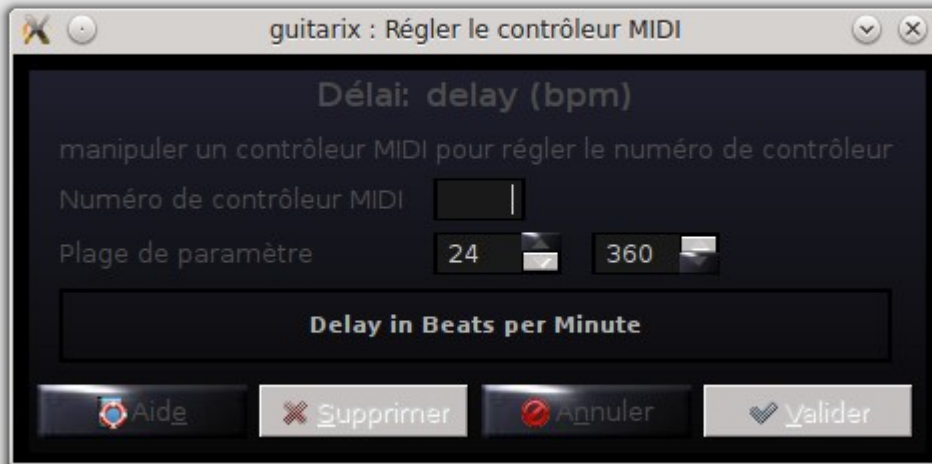
Second step:
Midi tab:

We connect the system system
output: MIDI_CAPTE_1 to the
system system output:
MIDI_PLAYBACK_1



Control via Midi

MIDI / Guitarix association



A click on the guitarix button to control with the middle of the mouse button or on the two buttons for a two buttons mouse reveals this dialog window.

Then you have to move the MIDI button you want to use. The references of this MIDI button are then recovered by guitarix. You have to click on "Validate" to record the Association Midi / Guitarix button.

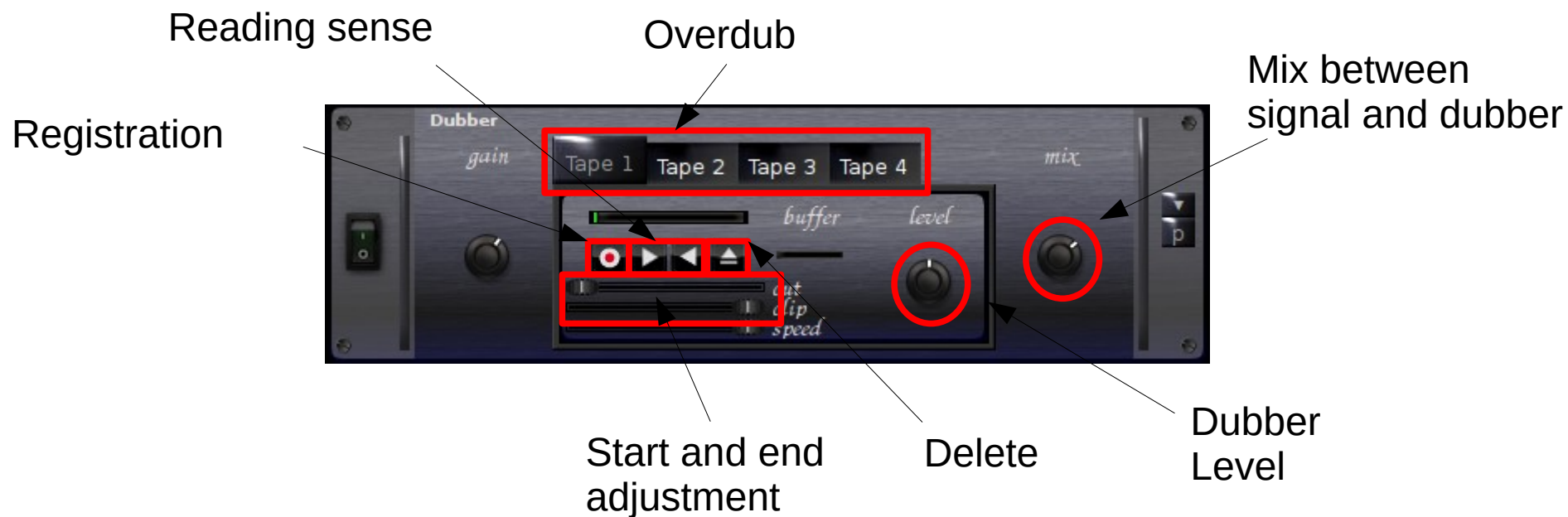


The engine menu → MIDI controller reveals this dialogue.

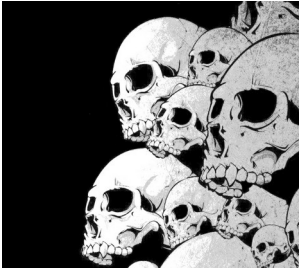
By clicking on "Including MIDI controllers in the Presets" allows you to save MIDI / Guitarix associations when you save the presets.



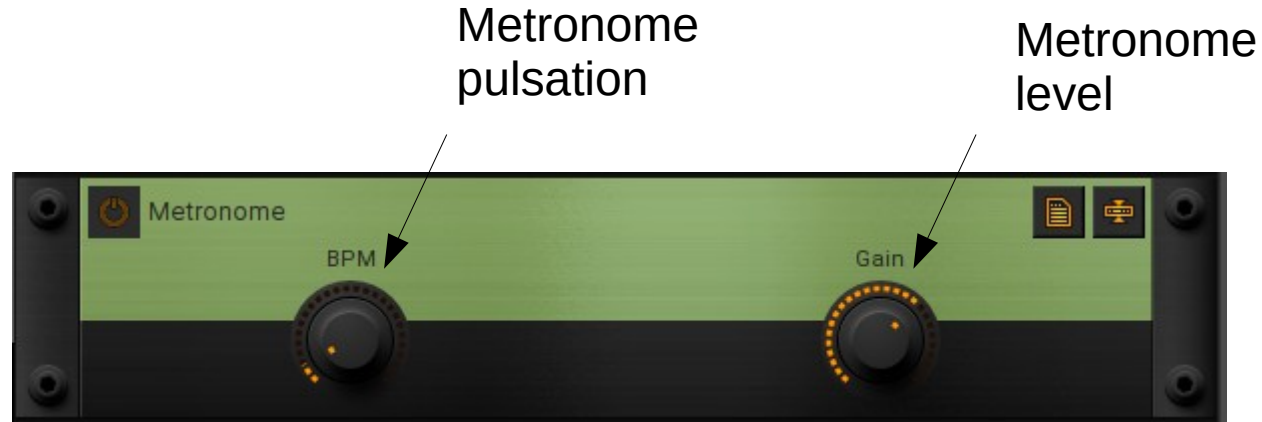
The Dubber



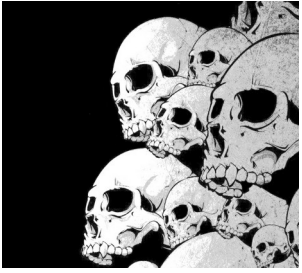
The Dubber allows you to record a sound loop and to replay it



The metronome



The metronome diffuses a beat. The sound level of the beat is proportional to the level of the sound at the input.



Guitarix and web server 1/2

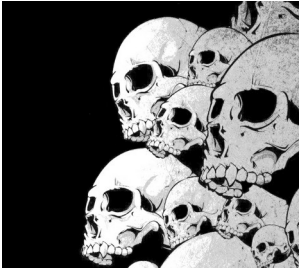
Guitarix can be launched without graphical interface.

It is then controllable via a web server.

This web server can then be checked via a graphics tablet for example.

Guitarix compilation in web server mode:

```
# We recover the source code of guitarix (Git must be installed)
$ git clone https://github.com/brummer10/guitarix
# We initialize guitarix dependencies
$ git submodule update --init --recursive
$ cd trunk/webui/bootplate/
$ git submodule update -init --recursive
$ cd ../ ..
# We are now compiling guitarix (GCC must be installed)
$ ./waf configure -prefix =/opt/guitarix-Devel
$ ./waf
# We install everything
$ su -c ./waf install
```



Guitarix and web server 2/2

1. Starting guitarix:

```
$ /opt/guitarix-devel/bin/guitarix -N
```

2. start the websockify program from the current directory (webui):

```
$ python -m websockify.websocketproxy --web=. '*' :8000 localhost:7000
```

3. open the URL <http://localhost:8000> in the desktop browser

You should be redirected to debug.html. index.html expects preprocessed files which can be generated with the deploy script:

```
$ ./deploy.sh
```

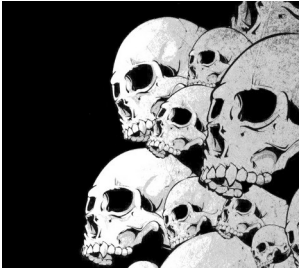
This script creates the build/ directory which is used in index.html.

deploy.sh needs the node program from node.js. For Debian/Ubuntu:

```
$ su -c dnf install nodejs
```

Next step :

- open the port 8000 on the tablet and on the server ;
- connect to the guitarix web server via the tablet ;
- install this web server on the Raspberry Pi.

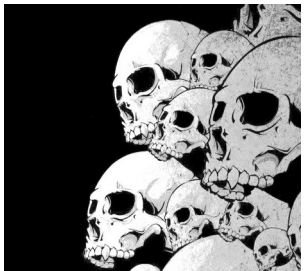


Guitarix Scope

The Scope allows you to visualize the signal to a given place in the rack. It is practical because it makes it possible to visualize the influence of the settings of an effect.

In this way, we can dose the quantity of distortion. Too much gain will generate an almost square signal. Not enough gain, the signal will remain sinusoidal. It remains to find the right dose.

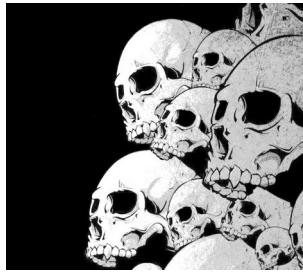




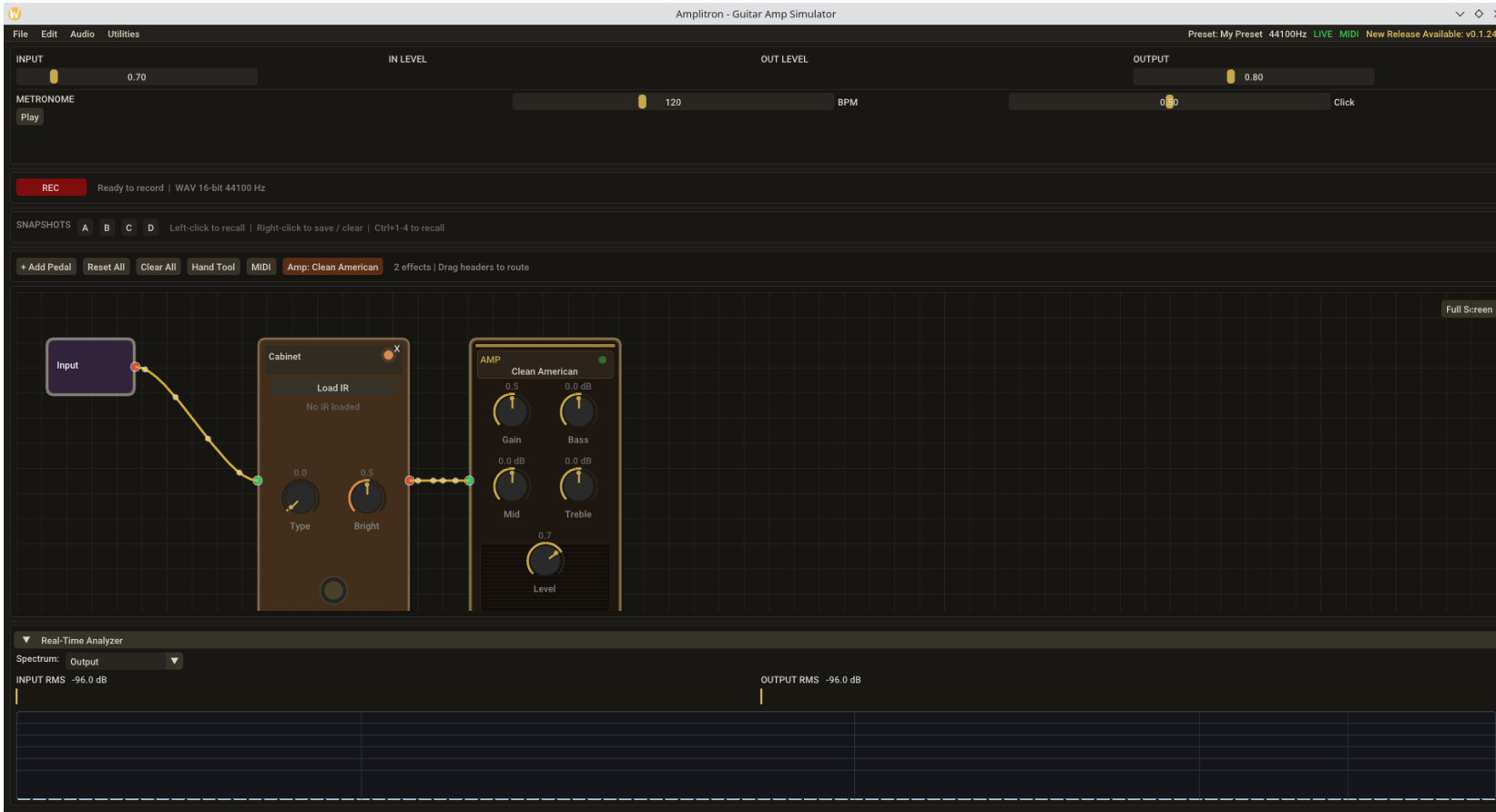
Guitarix / VST3

It's now possible to use
Guitarix as a VST3 plugin
in Carla for example



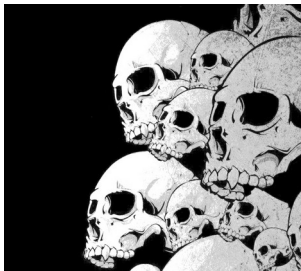


Amplitron

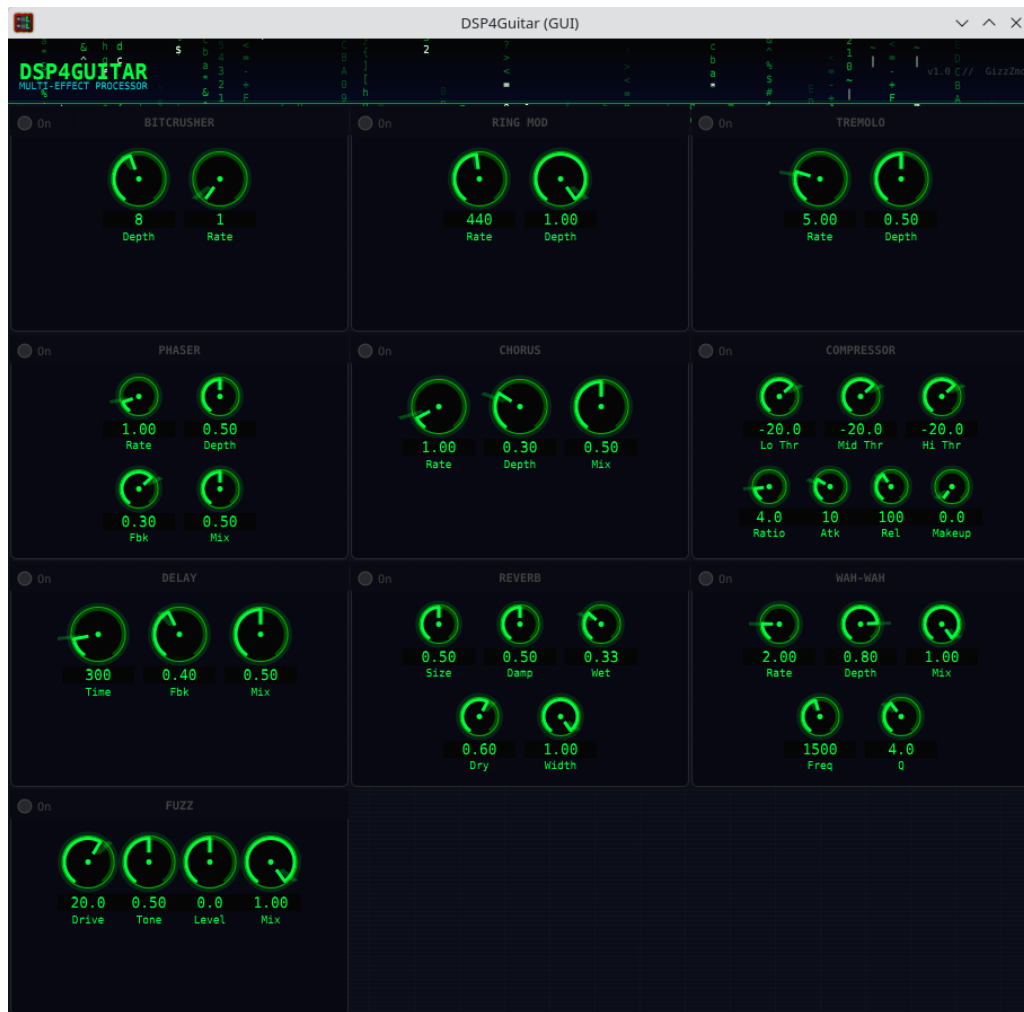


\$ dnf install amplitron

<https://github.com/sudip-mondal-2002/Amplitron>

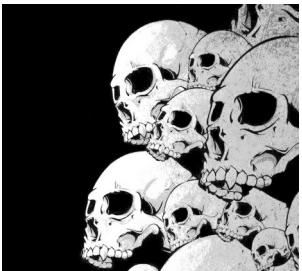


DSP4Guitar

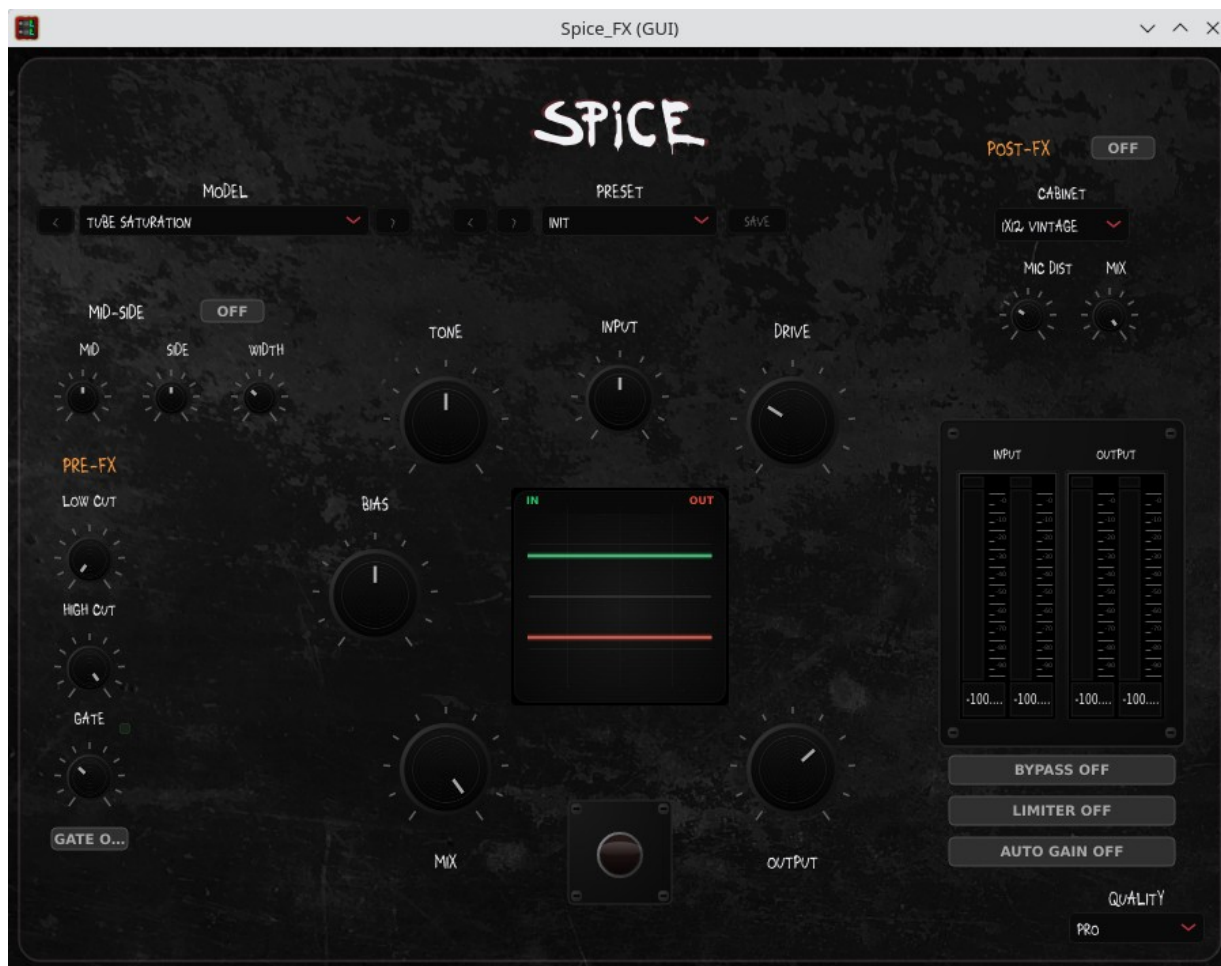


```
$ dnf install vst3-dsp4guitar
```

<https://github.com/GizzZmo/DSP4Guitar>

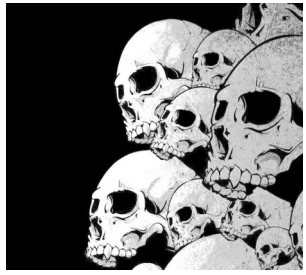


SpiceFX



```
$ dnf install lv2-spicefx  
$ dnf install vst3-spicefx
```

<https://github.com/DatanoiseTV/spice-oss>



Webography

You can access to a huge set of presets for Guitarix on Musical Artifacts :

<https://musical-artifacts.com/artifacts?apps=guitarix>

All these artifacts can be easily downloaded into Guitarix because Musical Artifacts is now integrated into Guitarix.



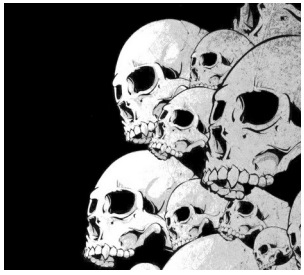
Documentary



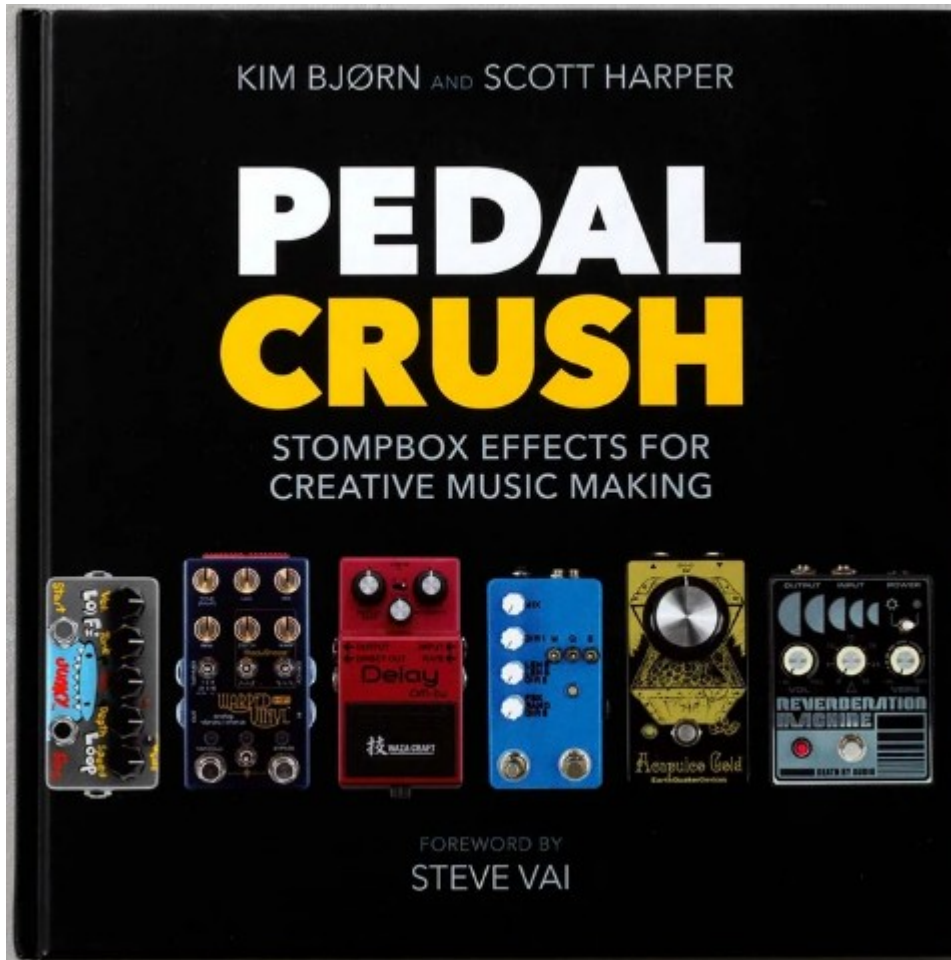
A good documentary about pedals, who made them ...

The PEDAL Movie :

<https://www.imdb.com/fr/title/tt11811310/>



Book



A good book about pedals :

PEDAL CRUSH

<https://bjooks.com/en-fr/products/pedal-crush-stompbox-effects-for-creative-music-making>